ArenaNeXt-World: Skill-Orchestrated Benchmarking for Interactive World Models

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Abstract

A world model maintains an internal state of an environment, updates that state under interventions, and renders observations from the evolving state. Existing benchmarks measure important signals such as visual quality, action following, navigation, physical plausibility, and memory, but separate scores over these abilities do not determine whether a model maintains one coherent world. A benchmark for world models should instead evaluate the full state cycle: whether the underlying world persists, whether interventions transform it correctly, and whether rendered observations remain reliable evidence of that world. We introduce **ArenaNeXt**, a next-generation evaluation paradigm that represents benchmark procedures as reusable, composable skills, and **ArenaNeXt-World**, its first domain instantiation for interactive world models. ArenaNeXt-World organizes evaluation from the state core outward: **World Persistence** tests whether the generated world remains a continuous, evolving state across time, occlusion, revisits, and viewpoints; **Transition Correctness** tests whether interventions induce intended state changes while preserving irrelevant variables; and **Observation Quality** tests whether rendered outputs provide reliable evidence for continued interaction. ArenaNeXt-World evaluates these axes through *WorldProbe*, an evaluation harness that instantiates axis-specific tests for different case types and calls metric skills to score the relevant evidence under the corresponding persistence, transition, or observation criterion. This skill design keeps diverse measurements tied to the world-state question they answer, while allowing ArenaNeXt-World to evolve within the broader ArenaNeXt series: new interactions, evidence sources, metric skills, and validators can be added as model interfaces and failure modes change, without changing the state-centered taxonomy or breaking axis-level comparability. The current release contains 228 interaction cases and nine evaluation skills, with 127 scored rollouts from five representative models. ArenaNeXt-World therefore evaluates current world models by the integrity of the worlds they maintain, transform, and render through interaction.

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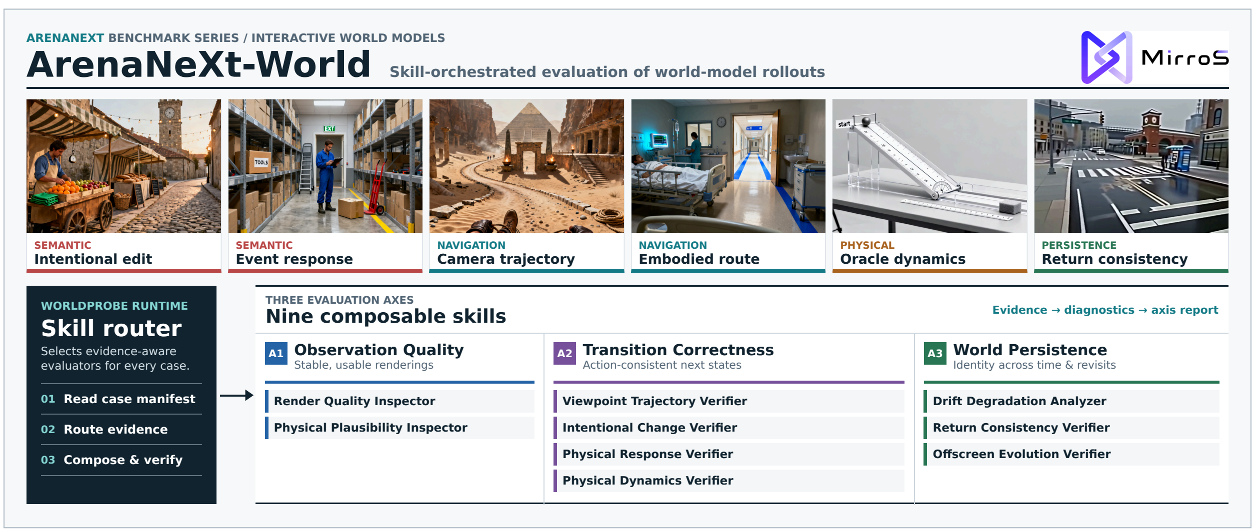

Figure 1: **ArenaNeXt-World.** The first world-model instantiation of ArenaNeXt routes diverse semantic, navigation, physical, and persistence cases to composable evaluation skills.

Table 1: Coverage comparison with recent video and world-model benchmarks, aligned to the same WorldProbe probe-family indicators used by the main experimental table. ✓ indicates explicit coverage with dedicated cases and metrics, △ indicates related but partial coverage, × indicates no explicit coverage, and – indicates not applicable or not reported.

	Observation Quality		Transition Correctness			World Persistence			Scale	
Benchmark	**Render (Obs-R)**	**Physical (Obs-P)**	**Exploratory (Exp-T)**	**Intentional (Int-T)**	**Physical (Phy-T)**	**Drift Resistance**	**Return / Revisit**	**Offscreen Evolution**	**Cases**	**Actions / Turns**
Video, generation, and interactive-world benchmarks										
VBench [1]	✓	×	×	×	×	×	×	×	946	946
WorldScore [2]	✓	×	✓	✓	×	△	×	×	3,000	3,000
WorldModelBench [3]	△	✓	✓	✓	△	×	×	×	350	350
MIND [4]	✓	×	✓	×	×	△	✓	△	250	–
WorldArena [5]	✓	✓	✓	△	△	✓	△	△	500	500
Omni-WorldBench [6]	✓	✓	✓	✓	△	△	✓	△	1,068	1,068
WorldLens [7]	✓	✓	✓	△	△	△	△	△	26k	–
WorldMark [8]	✓	×	✓	×	×	△	△	×	500	–
WBench [9]	✓	✓	✓	✓	△	✓	✓	△	289	1,058
Memory, revisit, and diagnostic benchmarks										
MemoBench [10]	△	△	△	△	△	✓	✓	✓	360	–
WRBench [11]	△	△	✓	△	△	✓	✓	✓	–	–
Physion-Atlas [12]	△	✓	×	△	✓	△	△	△	–	–
ArenaNeXt-World (ours)	✓	✓	✓	✓	✓	✓	✓	✓	**228**	**320**

Notes. The columns follow the same indicators as the main reporting table. Render covers perceptual, temporal, and readability checks; Physical Observation covers visible structural and physical plausibility. Exploratory Transition covers navigation, camera, and observation-only access changes; Intentional Transition covers entity, attribute, relation, and event changes; Physical Transition covers intervention-driven dynamics and counterfactual physical responses. The persistence columns cover long-horizon drift, leave-and-return consistency, and temporarily offscreen evolution. Scale values follow each benchmark's published unit; the ArenaNeXt-World release contains 228 cases and 320 controlled actions.

1 Introduction

Interactive world models are becoming a central interface between generative modeling and interactive intelligence. Unlike conventional video generation, which primarily synthesizes a plausible observation sequence, an interactive world model is expected to maintain a latent world state, update that state under interventions, and render future observations that remain grounded in the same evolving world. This requirement is fundamental to embodied agents, game engines, simulation systems, robot planning, and interactive content generation, where users or agents continuously observe, act, and revise their decisions based on the consequences of previous actions. Evaluation should therefore assess not only whether a generated clip appears natural, but whether the model renders reliable observations, evolves correctly under actions, and preserves world identity over time.

This evaluation problem is intrinsically procedural. Different interaction cases require different evidence: video-quality metrics for perceptual fidelity, optical flow and temporal statistics for motion stability, tracking and segmentation for object identity, counterfactual rollouts for action specificity, simulator or analytic evidence for physical consistency, structured VLM judging for semantic events, and human calibration for preference-aligned validation. A benchmark for interactive world models must therefore specify how to select evidence sources, invoke metric modules, validate measurements, and aggregate heterogeneous signals. Treating these procedures as an unstructured collection of scorers obscures when each metric is applicable, what evidence supports it, and how failures should be diagnosed.

Existing benchmarks provide important ingredients for this goal, but their evaluation protocols remain fragmented. Video-generation benchmarks emphasize perceptual quality, temporal coherence, prompt alignment, and human preference; interactive and world-model benchmarks add camera control, navigation, action following, physical plausibility, or memory-style probes; persistence benchmarks further test whether objects or scenes can be recovered after disappearance or return. However, these protocols are typically instantiated as fixed metric taxonomies or task-specific pipelines. As a result, prior evaluation recipes are difficult to transfer across model interfaces, extend to new interaction types, or combine into a unified diagnostic view of world-model behavior. Table 1 summarizes this gap across the three evaluation capabilities used throughout this report: Observation Quality, Transition Correctness, and World Persistence.

We introduce **ArenaNeXt**, a next-generation skill-based evaluation paradigm designed as an extensible series, and **ArenaNeXt-World**, its first domain instantiation for interactive world models. ArenaNeXt reformulates benchmark evaluation as a reusable *evaluation skill library*. Each evaluation skill specifies an applicability condition, required evidence, scoring module, validation criterion, and aggregation rule. Under this formulation, previous benchmark procedures can be absorbed as skills rather than duplicated as isolated pipelines: video-quality assessment becomes a family of observation-evaluation skills, action-following checks become transition-evaluation skills, memory and revisit probes become persistence-evaluation skills, and structured VLM or human-alignment protocols become validation skills. This skill-based representation makes benchmark construction compositional, auditable, and continuously extensible as new model capabilities, evidence sources, and failure modes emerge.

Prior world-model literature provides the modeling basis for this organization. Latent world models represent an environment through hidden states that are updated by actions and decoded into observations [13–15]. Recent generative interactive environments and game-style world models expose the same stateful rollout problem through visual observations, camera or control actions, and generated future frames [16–20]. Evaluation work has also converged on complementary evidence types: video benchmarks measure rendered observation quality [1, 21, 22]; world-model benchmarks evaluate action following, controllability, physical response, and embodied utility [3, 2, 8, 23, 6, 5, 7, 12]; and memory-oriented benchmarks test whether a persistent world can be recovered across disappearance, revisits, and multi-turn rollouts [9–11, 4]. ArenaNeXt-World therefore uses the following observation-conditioned factorization as an evaluation view of interactive world modeling:

$$\boldsymbol{o_i}, \ldots, \boldsymbol{o_t} \propto S(\boldsymbol{o_t} \mid \boldsymbol{s_t})\, T(\boldsymbol{s_t} \mid \boldsymbol{s_{t-1}}, \boldsymbol{a_{t-1}})\, \Pi_{i<t} P(\boldsymbol{o_i} \mid \boldsymbol{o_t})$$

The proportionality is not meant to introduce a normalized likelihood for every model interface. Instead, it separates the observable evidence required to evaluate a rollout. The current observation must be a usable rendering of the current latent world state; the current latent state must be a correct consequence of the previous state and action; and earlier observations must remain compatible with the same world identity when compared against the current observation. These three terms induce the three ArenaNeXt-World axes. **Observation Quality** evaluates $S(\boldsymbol{o_t} \mid \boldsymbol{s_t})$, asking whether rendered evidence is visually stable, physically plausible, and usable for subsequent interaction. **Transition Correctness** evaluates $T(\boldsymbol{s_t} \mid \boldsymbol{s_{t-1}}, \boldsymbol{a_{t-1}})$, asking whether a controlled action induces the expected next state while preserving unrelated variables. **World Persistence** evaluates $\Pi_{i<t} P(\boldsymbol{o_i} \mid \boldsymbol{o_t})$, asking whether earlier observations remain consistent with the same world identity through long rollouts, revisits, offscreen disappearance, and return. These axes define what ArenaNeXt-World evaluates, while the evaluation skills define how each axis is operationalized for a specific case and evidence configuration.

ArenaNeXt-World contains **228** interaction cases—90 semantic, 55 navigation, 59 physical, and 24 persistence cases—with **320** controlled actions and **9** evaluation skills spanning Observation Quality, Transition Correctness, and World Persistence. We report **127** scored rollouts from **5** representative world models across text-controlled, camera-controlled, and action-conditioned interfaces, with a further 20 models in the evaluation queue. The resulting analysis distinguishes perceptual plausibility from reliable world modeling: a model may generate high-quality videos while failing to produce correct action consequences, physically grounded transitions, or persistent world identity. ArenaNeXt-World therefore contributes a concrete world-model benchmark while establishing ArenaNeXt as a general skill-based methodology for unifying, extending, and diagnosing evaluation across future domain-specific instantiations.

2 Related Works

2.1 Video Generation Benchmarks

Early and mainstream video generation benchmarks mainly start from the video-benchmark perspective and aim to measure the quality of generated videos themselves. Representative efforts such as VBench [1] and VBench++ [21] decompose *video generation quality* into fine-grained dimensions, including image quality, aesthetics, motion smoothness, temporal flickering, subject consistency, and prompt alignment, and they improve evaluation reliability through both automatic metrics and human-preference validation. VBench-2.0 [22] pushes this line further from *superficial faithfulness* toward *intrinsic faithfulness* by introducing dimensions such as physics, commonsense, controllability, and related fine-grained capabilities. These benchmarks provide important quality standards for video generation models, but their central object is still whether a generated clip is natural, credible, and conditionally aligned. In other words, they primarily evaluate visual observation, rather than how a world continues to evolve under actions.

2.2 Observation-Centric World Model Benchmarks

With the rise of world models and interactive world models, a number of benchmarks have begun to introduce camera control, trajectory following, viewpoint change, action following, or memory-style tasks. For example, iWorld-Bench [23] constructs a unified action-generation framework and evaluates world models from the perspectives of visual generation, trajectory following, and memory. These benchmarks are closer to interactive world modeling than traditional video benchmarks, but much of the so-called interaction still lives at the observation level: the model receives a camera, view, or trajectory action and produces a new visual observation, while the primary evaluation focus remains whether the view follows the action, whether the frame looks reasonable, and whether the trajectory is continuous. Next-generation evaluation should not stop at observation-level interaction. A benchmark for truly interactive world models should jointly test observation changes, event changes, and physical actions, as well as whether a world state continues to evolve consistently under those actions.

2.3 Memory and Persistence Benchmarks

Another line of work focuses on whether world models possess memory and persistence. MemoBench [10] adopts a Visible–Disappear–Reappear protocol to test whether a target object can be correctly recovered after leaving the field of view, undergoing dynamic changes, and then reappearing. This directly touches the core question of world persistence: whether the model can preserve object identity, geometry, and state evolution during periods of invisibility. We view these memory benchmarks as important, but they cover only one local slice of persistence, mainly object permanence or offscreen reappearance. In an interactive world model, persistence means more than remembering an object after it disappears. The model must maintain the same world through multi-round actions and long-horizon evolution: world quality should not keep degrading over time, the environment and objects should remain consistent when the agent leaves and returns to the same place, offscreen events should be reflected correctly when they become visible again, and each transition should produce a stable next state that can serve as the basis for later evolution.

3 A State-Centered Taxonomy for World Models

3.1 Formalizing a World Model

A world model maintains an internal state of an environment, updates that state under interventions, and renders observations from the evolving state. ArenaNeXt-World formalizes this state cycle through an observation-conditioned factorization:

$$\boxed{\boldsymbol{o_i}, \ldots, \boldsymbol{o_t} \propto S(\boldsymbol{o_t} \mid \boldsymbol{s_t})\, T(\boldsymbol{s_t} \mid \boldsymbol{s_{t-1}}, \boldsymbol{a_{t-1}})\, \Pi_{i<t} P(\boldsymbol{o_i} \mid \boldsymbol{o_t})} \tag{1}$$

Here $S(\boldsymbol{o_t} \mid \boldsymbol{s_t})$ denotes whether the current rendered observation is reliable evidence of the latent world state. $T(\boldsymbol{s_t} \mid \boldsymbol{s_{t-1}}, \boldsymbol{a_{t-1}})$ denotes whether the previous intervention induces the correct current state. The persistence term $\Pi_{i<t} P(\boldsymbol{o_i} \mid \boldsymbol{o_t})$ does not require earlier observations to be reproduced pixel-wise; it asks whether earlier observations remain explainable as views, traces, or commitments of the same continuous world. This factorization induces ArenaNeXt-World's three evaluation axes: **Observation Quality**, **Transition Correctness**, and **World Persistence**.

3.2 Three Evaluation Axes

ArenaNeXt-World orders the axes from rendered evidence to interaction-conditioned state updates and then to longer-horizon world continuity. This ordering matches the evaluation path: a benchmark must first verify that observations are usable evidence, then test whether interventions transform the state correctly, and finally ask whether the same world persists across time, occlusion, revisits, and viewpoints.

Observation Quality. Observation Quality tests whether rendered outputs provide reliable evidence of the current world state for continued interaction. This axis includes perceptual quality, physical and structural plausibility, temporal coherence, and the readability of state-relevant entities, attributes, relations, and events. It is not merely a video preference score: an observation can look attractive while hiding the target object, making the action consequence ambiguous, or rendering a scene that cannot support later state checks.

Transition Correctness. Transition Correctness tests whether interventions induce intended state changes while preserving irrelevant variables. The benchmark asks whether the model updates the right part of the world, in the right direction, under the right constraints. This goes beyond prompt adherence or endpoint plausibility: a model should not receive full credit for producing a plausible future if the future is insensitive to the specified intervention, changes the wrong entity, or damages unrelated state.

Axis	Probe Family	Question
Observation Quality	Render	Is the rendered observation usable evidence of the current world state?
	Physical	Is the rendered scene structurally and physically reasonable?
Transition Correctness	Exploratory Transition	Does observation-only exploration update the view while preserving the world?
	Intentional Transition	Do agent actions change intended entities or relations?
	Physical Transition	Do changed physical conditions induce correct evolution?
World Persistence	Drift Resistance	Do invariant commitments avoid long-horizon drift?
	Return / Revisit Consistency	Does the world remain compatible after leaving and returning?
	Offscreen Evolution	Do endogenous dynamics continue rather than freeze or reset?

Table 2: ArenaNeXt-World taxonomy and probe families. The top-level axes are stable world-state commitments; probe families instantiate those commitments in concrete interaction settings.

World Persistence. World Persistence tests whether the generated world remains a continuous, evolving state beyond the current observation. The benchmark asks whether the same world survives long rollout, temporary absence from view, revisits to earlier places, and viewpoint changes. Persistence does not mean freezing the scene: invariant commitments should remain stable, while endogenous processes should continue to evolve when the world dynamics imply that they should.

3.3 Probe Families

The three axes define the state commitments that a world model should satisfy; they are not task categories. ArenaNeXt-World instantiates each axis through probe families that target recurring failure pressures in generated worlds. A single case may activate multiple families: for example, a navigation rollout can test exploratory transition, render evidence quality, and return consistency.

Observation probes. Observation probes evaluate the reliability of rendered evidence. *Render* checks whether the observation is perceptually coherent, temporally stable, and free from severe artifacts such as collapse, duplication, local distortion, or unstable appearance. It also includes evidence readability: entities, relations, and state variables needed by later evaluation must be visible and inspectable. A visually impressive rollout can still fail this probe if it hides the evidence needed to judge the world. *Physical* checks whether the rendered scene itself respects basic structural and physical constraints, including plausible support, contact, occlusion, scale, spatial arrangement, and object integrity.

Transition probes. Transition probes evaluate intervention-conditioned state updates. *Exploratory Transitions* change only the observer's access to the world, such as looking around, moving forward, turning, following a camera path, or navigating to a different pose; the underlying scene should remain compatible as the view changes. *Intentional Transitions* change entities, attributes, relations, or events through agent actions, such as opening, moving, placing, removing, or triggering an object. *Physical Transitions* change dynamical conditions or physical parameters and test whether the resulting evolution follows the corresponding constraints. In this sense, physical in Observation concerns whether the rendered state is reasonable, while physical in Transition concerns whether an intervention causes the correct dynamical consequence.

Persistence probes. Persistence probes evaluate continuity of the generated world. *Drift Resistance* tests whether invariant commitments such as identity, attributes, relations, and layout remain stable over long rollouts. *Return / Revisit Consistency* tests whether state that leaves the current view remains geometrically and semantically compatible when the camera returns, an occluder moves away, or the agent re-enters a previously observed place. *Offscreen Evolution* tests whether endogenous dynamics continue during elapsed time or temporary invisibility rather than freezing,

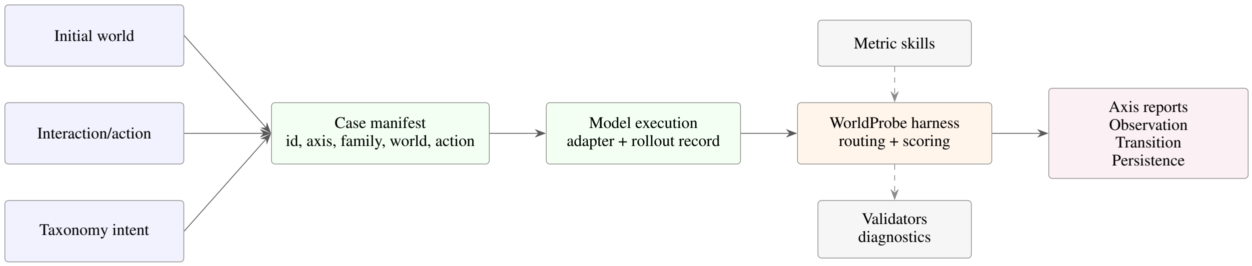

Figure 2: **WorldProbe overview.** Within ArenaNeXt-World, diverse initial worlds and interactions become cases whose rollouts are routed through the skill-orchestrated evaluation harness and reported as factorized evidence for Observation Quality, Transition Correctness, and World Persistence.

resetting, or jumping arbitrarily; examples include dripping water, burning fire, moving fluids, or other processes that should keep evolving without a new external action.

3.4 From Taxonomy to WorldProbe

The taxonomy defines what question each score answers. It does not prescribe one universal metric for every case, because different interaction settings expose different evidence. Within ArenaNeXt-World, *WorldProbe* operationalizes the taxonomy as an evaluation harness that instantiates a case, collects the required observations and traces, selects applicable metric skills, validates the measurement, and aggregates the result into the corresponding axis.

Under this design, probe families and metric skills live below the stable taxonomy. New interactions, evidence sources, model interfaces, metric skills, and validators can be added as world models and their failure modes evolve, while the top-level Observation–Transition–Persistence structure remains comparable across benchmark versions.

4 WorldProbe: Executable Evaluation

The taxonomy in Eq. (1) defines the state-cycle questions that ArenaNeXt-World asks. *WorldProbe* defines how these questions are executed. An ArenaNeXt-World case is a controlled world-state query: it specifies the world a model enters, the action or observation request applied to that world, and the evaluation intent that tells the harness which axis and probe family the rollout should answer. The remaining details–how a request is adapted to a model interface, what rollout evidence is collected, which metric skills are applicable, which measurements are invalid, and how scores are aggregated–belong to the WorldProbe runtime rather than to the case itself.

This separation is important for an evolving benchmark. Case construction should remain stable as model interfaces change from text-to-video and image-to-video to camera control, action-conditioned rollout, or interactive APIs. Conversely, the runtime should be able to add new adapters, validators, evidence extractors, and metric skills without rewriting the case manifest. ArenaNeXt-World therefore represents a case by a small core while the runtime supplies the execution and measurement contract.

4.1 Case Construction

ArenaNeXt-World constructs each case by pairing an initial world with an interaction and by assigning the evaluation intent that tells WorldProbe how the resulting rollout should be judged. In this sense, case construction is *initial world plus interaction plus evaluation intent*, not prompt enumeration.

Initial observation construction. The initial observation is the evidence through which a model enters a world. It may be a generated image, a simulator first frame, a rendered physical setup, or, in later versions, a real frame or short video context. We construct it through a four-step pipeline:

$$\text{coverage seed} \rightarrow \text{world specification} \rightarrow \text{initial-observation materialization} \rightarrow \text{filtering and validation.}$$

Coverage seeds control diversity in scenes, viewpoints, subjects, domains, and visual styles. They include indoor rooms, outdoor routes, urban intersections, natural environments, game-like navigation spaces, and controlled physical systems; they also cover first-person, third-person, fixed-camera, and measurement-camera views. The goal is not scene variety for its own sake, but coverage of the state conditions under which world models must be evaluated.

Each seed is expanded into a world specification. For exploratory-transition cases, the specification identifies traversable routes, look targets, occlusion-safe landmarks, and observation affordances. For intentional transition cases, it identifies editable entities, attributes, relations, local events, entry regions, and stable anchors that should survive unrelated actions. For physical cases, it identifies measurement-friendly objects, controllable parameters such as force, angle, mass, or friction, fixed calibration anchors, and measurable state variables such as trajectory, contact order, collision count, fill height, or final position. For persistence cases, it emphasizes stable anchors and endogenous dynamics, such as dripping water, flames, flowing fluids, dust, or melting objects.

The world specification is then materialized into an initial observation. Most semantic and navigation cases use generated head frames selected from candidate images. Physical cases additionally use simulator or renderer outputs: the first frame defines the initial state, while control videos or trajectories provide reference evidence for later metric skills. This design lets ArenaNeXt-World mix natural visual worlds with oracle-backed physical systems under the same case abstraction.

Finally, cases are filtered before release. The initial observation must make the relevant target, anchors, route, or physical setup visible; it must not already contain the future action outcome; and it must be compatible with the intended probe family. For example, a revisit case needs recognizable anchors, a physical case needs a view from which state variables can be measured, and an offscreen-evolution case needs an endogenous process that can plausibly continue while temporarily unseen.

Interaction construction. After the initial world is fixed, ArenaNeXt-World derives an action from two sources: the affordances of the world and the template of the probe family. This prevents actions from being arbitrary prompts detached from the visual state. The construction follows

$$\text{world affordances} \rightarrow \text{probe-family template} \rightarrow \text{concrete action} \rightarrow \text{optional variants or counterfactuals.}$$

Exploratory-transition actions are observation-only transitions: they change the model's access to the world without intentionally editing entities, relations, or physical parameters. They are drawn from navigation and observation affordances such as camera paths, WASD-style movement, look-around commands, route following, and the move-away segment of a later return probe. Intentional-transition actions are drawn from controllable entities and relations, such as opening a book, changing an object attribute, moving an item, triggering a local event, or introducing an agent through a visible entry region. Physical-transition actions are drawn from structured physical knobs and typically form parameter sweeps or counterfactual groups. Persistence actions are drawn from anchors and endogenous dynamics: a drift probe requests a long rollout, a return/revisit probe moves away and returns, and an offscreen-evolution probe hides a process and later observes whether it continued rather than froze or reset.

The action may be a simple text instruction, a discrete control sequence, a camera trajectory, a multi-chunk rollout plan, or a physical parameter set. Rollout length is therefore part of the action when it is case-specific. For instance, an intentional object-state change may use a short rollout, while a drift probe may define several long chunks and a revisit probe may define turn-away, wait-or-move, and return segments. If the action does not specify a horizon, the probe-family configuration provides a default.

Minimal case manifest. The release manifest stores only the stable core of the query:

Field	Role
`case_id`	unique case identifier
`taxonomy.primary_axis`	one of Observation, Transition, Persistence
`taxonomy.probe_family`	concrete family such as exploratory transition or drift resistance
`world.initial_observation`	image, video context, simulator frame, or other entry evidence
`interaction.action`	text action, control sequence, camera path, chunk plan, or physical sweep

This minimal core keeps the case independent of a particular model interface or metric implementation. The model adapter receives the initial observation and action; the WorldProbe harness additionally uses the taxonomy fields for skill planning, diagnostics, and axis-level reporting. Fields used during construction–world specifications, candidate anchors, expected outcomes, protected regions, negative constraints, and review notes–remain useful for generating and auditing cases, but they need not be mandatory fields in the release manifest. When a family requires extra structure, such as a physical parameter sweep, a simulator reference, or a multi-step revisit path, that information is encoded inside `interaction.action` or in the runtime configuration rather than as a new global requirement for every case.

4.2 WorldProbe Protocol

Case construction defines what world-state query should be asked. The WorldProbe protocol defines how that query is executed against a model and how the resulting evidence is evaluated. Its design has three components: adapters convert a case into model-specific inputs, evidence records standardize the rollout returned by the model, and a skill library supplies the evaluation agents used for scoring.

4.2.1 Adapters and Evidence Records

A model adapter receives the model-facing part of a case: the initial observation and the interaction action. It maps this pair into the interface required by a particular model, such as a text-to-video prompt, an image-to-video request, a camera trajectory, a discrete control sequence, a video context, or an interactive API call. The taxonomy fields are not treated as model instructions by default; they are used by the WorldProbe harness to interpret the rollout and select evaluation skills. This keeps the same case comparable across models without exposing the evaluation target as an extra hint to the generator.

After execution, WorldProbe converts the model output into a rollout evidence record r. The record contains generated frames or videos, action metadata, adapter metadata, timestamps or chunk boundaries, and any optional traces required by the case, such as camera paths, object tracks, simulator references, counterfactual pairs, or recovered physical variables. The evidence record is larger than the case manifest because it is a runtime artifact: it stores what was actually produced and what can be measured, while the case manifest stores the stable query.

4.2.2 Skill Library and Routing

WorldProbe evaluates rollouts through a release skill library $\mathcal{L}$. A skill is an evaluator agent registered with metadata, applicability conditions, required evidence, an evaluation procedure, and a structured output schema:

$$\sigma_j = (m_j, g_j, E_j, P_j, O_j). \tag{2}$$

Here m_j describes the skill and its intended scope, g_j decides whether the skill applies to a case and rollout, E_j specifies the required evidence, P_j is the evaluation procedure, and O_j defines the returned score, confidence, evidence span, and diagnostic tags. A skill may be implemented by deterministic rules, visual or language model judgement, tracking, camera recovery, simulation or analytic oracles, physical parsers, or combinations of these tools. The benchmark fixes the registered skills and output schemas for a release; individual cases do not invent new metrics.

Given a case c, rollout evidence record r, and skill library $\mathcal{L}$, WorldProbe routes evaluation by selecting applicable skills:

$$\mathcal{S}(c, r) = \{\sigma_j \in \mathcal{L} \mid g_j(c, r) = 1\}. \tag{3}$$

Routing uses the taxonomy intent, probe family, action structure, rollout evidence, and skill metadata. The selected skills then execute on the evidence record and return structured outputs; if required evidence is missing, the output records the missing evidence rather than silently assigning an unreliable score. This makes the protocol flexible enough to support heterogeneous measurements while keeping leaderboard configuration reproducible.

The routing matrix in Table 3 summarizes how the release skill agents serve the state-centered probe families. A check mark means that a skill can be selected for that family when the case and rollout provide the required evidence; an empty entry means the skill is not routed for that family. Render quality is treated as a global evidence gate: outside the Observation–Render family, it is used mainly to validate whether subsequent transition or persistence measurements are judgeable, rather than to replace the family-specific score. The detailed metric composition of each skill is documented separately in the release skill cards, so the main text can emphasize the taxonomy-level routing rather than a flat list of raw metrics.

4.2.3 Scoring and Reporting

Each selected skill returns a normalized score together with confidence and diagnostics. WorldProbe first summarizes the skill outputs into a case-level profile, then aggregates cases within probe families, and finally reports the three taxonomy axes: Observation Quality, Transition Correctness, and World Persistence. The primary leaderboard view is this factorized profile because it shows whether a model fails at rendering reliable observations, transforming state under intervention, or maintaining a continuous evolving world. A single overall score can be reported as a secondary summary, but it is not the main object of analysis.

Skill Agent	Obs-R	Obs-P	Exp-T	Int-T	Phy-T	Drift	Return	Offscreen
RENDERQUALITYINSPECTOR	✓	✓	✓	✓	✓	✓	✓	✓
PHYSICALPLAUSIBILITYINSPECTOR		✓			✓			
VIEWPOINTTRAJECTORYVERIFIER			✓				✓	
INTENTIONALCHANGEVERIFIER				✓				
PHYSICALRESPONSEVERIFIER					✓			
PHYSICALDYNAMICSVERIFIER					✓			✓
DRIFTDEGRADATIONANALYZER	✓					✓		
RETURNCONSISTENCYVERIFIER			✓				✓	
OFFSCREENEVOLUTIONVERIFIER					✓			✓

Table 3: WorldProbe skill-to-taxonomy routing matrix. Obs-R and Obs-P denote the Render and Physical probe families under Observation Quality; Exp-T, Int-T, and Phy-T denote Exploratory, Intentional, and Physical Transition. Detailed metrics and implementation checks are specified in release skill cards.

5 Experiments and Analysis

This release reports the first ArenaNeXt-World MiniBench v0 snapshot: 127 scored rollouts from five representative models across the probe families supported by their current interfaces. The table reports each available family score separately before aggregation, so a strong rendering score cannot conceal weak action-conditioned transition correctness or persistence. Unevaluated model–family pairs remain explicit rather than being imputed.

The scored snapshot is accompanied by a 20-model evaluation queue spanning text-driven, camera-controlled, and action-conditioned systems. This roster makes the next coverage targets explicit while preserving a clear boundary between released scores and pending runs. Subsequent evaluation rounds will fill missing probe-family cells and extend the analysis with action-family breakdowns, category-level failure profiles, correlations against video-quality and preference-style scores, and ablations of counterfactual pairs, perception front-ends, and persistence sweep lengths.

Model	Observation Quality		Transition Correctness			World Persistence			Avg.	Status
	Render	Physical	Explor.	Intent.	Physical	Drift	Return	Offscreen		
Scored MiniBench v0 snapshot										
Wan2.2-I2V-A14B	0.843	0.968	0.771	1.000	0.544	–	–	–	0.825	scored
LingBot-World v2	0.860	0.936	0.847	0.767	0.531	0.984	0.799	–	0.818	scored
Lyra2	0.831	0.960	0.844	0.867	0.503	–	–	–	0.801	scored
Wan2.2-TI2V-5B	0.833	0.968	0.848	0.733	0.531	–	–	–	0.783	scored
HunyuanVideo 1.5 I2V	0.854	0.976	0.919	0.867	0.522	0.929	0.740	–	0.830	scored
Pending text-driven models										
Seedance 1.5	–	–	–	–	–	–	–	–	–	pending
Wan 2.7	–	–	–	–	–	–	–	–	–	pending
Kling 3.0	–	–	–	–	–	–	–	–	–	pending
YUME 1.5	–	–	–	–	–	–	–	–	–	pending
LTX 2.3	–	–	–	–	–	–	–	–	–	pending
LongCat	–	–	–	–	–	–	–	–	–	pending
Kairos 3.0	–	–	–	–	–	–	–	–	–	pending
Cosmos 2.5	–	–	–	–	–	–	–	–	–	pending
Pending camera-controlled models										
LingBot-World	–	–	–	–	–	–	–	–	–	pending
HY-World 1.5	–	–	–	–	–	–	–	–	–	pending
HY-World 2	–	–	–	–	–	–	–	–	–	pending
Fantasy-World	–	–	–	–	–	–	–	–	–	pending
InSpatio-World	–	–	–	–	–	–	–	–	–	pending
Astra	–	–	–	–	–	–	–	–	–	pending
Pending action-conditioned models										
Happy Oyster	–	–	–	–	–	–	–	–	–	pending
Matrix-Game 3.0	–	–	–	–	–	–	–	–	–	pending
Genie 3	–	–	–	–	–	–	–	–	–	pending
Matrix-Game 2.0	–	–	–	–	–	–	–	–	–	pending
HY-GameCraft	–	–	–	–	–	–	–	–	–	pending
Infinite-World	–	–	–	–	–	–	–	–	–	pending

Table 4: Current ArenaNeXt-World MiniBench v0 leaderboard and pending model roster. '–' indicates that the corresponding probe family has not yet been evaluated for that model in this run. Observation Quality columns are aggregated over all evaluated rollouts using RENDERQUALITYINSPECTOR and PHYSICALPLAUSIBILITYINSPECTOR.

6 Limitations

The initial release does not cover the full space of interactive worlds. Several limitations therefore qualify the current snapshot.

Domain coverage. The released cases cover a finite set of scene families and physical systems. Strong performance on these cases should not be interpreted as universal world modeling.

Simulation-real gap. Simulation provides valuable ground truth for physical intervention tests, but simulated systems may not capture all visual and material complexity of real scenes.

Measurement front-ends. Tracking, segmentation, depth, and camera recovery can fail. Reported scores therefore depend on front-end validation and explicit failure-handling rules.

Semantic ambiguity. Some semantic interventions do not have a single exact future. These cases should use structured invariants and counterfactual comparisons rather than unconstrained preference judgments.

7 Conclusion

ArenaNeXt-World is the first domain instantiation of ArenaNeXt, a next-generation skill-based evaluation paradigm, and reframes world-model evaluation around controlled interaction. The central target is not whether a generated clip is preferred, but whether a world state transitions correctly under an action and remains persistent as interaction continues. The benchmark instantiates this target through controlled initial states, action families, counterfactual pairs, verifiable measurements, and factorized reporting across Observation Quality, Transition Correctness, and World Persistence.

The v0 release realizes this design with 228 cases, 320 controlled actions, and nine evaluation skills. Its first snapshot reports 127 rollouts from five representative models and publishes a 20-model queue for broader interface coverage. Subsequent releases will complete missing probe-family cells, expand evidence panels for recovered state variables, and deepen model comparisons with category-level failure signatures alongside the factorized and aggregate scores.

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A WorldProbe Skill Library Details

This appendix expands the routing matrix in Table 3. The main paper reports which skill agents can be routed to each probe family; here we summarize the representative metrics and checks encapsulated by each agent. These metrics are implementation details of a release skill card, not additional taxonomy axes.

Each skill card has the same role in WorldProbe: it receives a rollout evidence record, checks whether its required evidence is available, computes one normalized score together with diagnostics, and reports whether the measurement should be trusted. This design keeps the case manifest lightweight. Cases specify the initial world, action, and taxonomy intent; the skill library specifies how observable evidence is measured for a particular release.

A.1 Skill Cards

Render Quality Inspector. This skill acts as both an Observation–Render evaluator and a global evidence gate. It checks whether the generated frames are perceptually coherent, temporally stable, and sufficiently readable for downstream scoring. Its diagnostics include aesthetic quality, imaging quality, temporal flicker, motion smoothness, visible artifacts, and whether target entities or state variables are inspectable. In non-render probe families, a low render-quality score marks later transition or persistence scores as low-confidence rather than replacing their family-specific measurements.

Physical Plausibility Inspector. This skill evaluates whether the rendered state is structurally and physically reasonable before asking whether an action succeeded. It checks support and contact, object integrity, scale and perspective, occlusion consistency, and whether visible motion is physically feasible. The output is useful for Observation–Physical cases and as a diagnostic for Physical Transition cases: a transition can appear active while still producing an impossible intermediate state.

Viewpoint Trajectory Verifier. This skill evaluates exploratory transitions. It checks whether the observer or camera motion implied by the action is reflected in the rollout, using evidence such as camera-path adherence, navigation consistency, layout recovery, and return-trigger support. It does not judge whether an object was intentionally edited. Its output is therefore routed to Exploratory Transition and can support Return/Revisit Consistency when a case includes a leave-and-return segment.

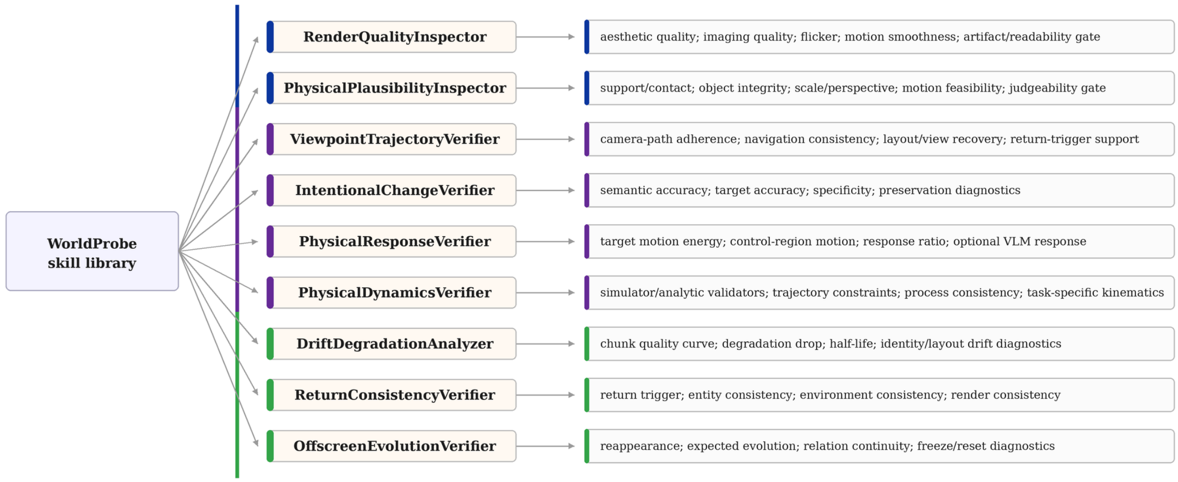

Figure 3: **Skill-to-metric expansion.** WorldProbe routes cases to reusable skill agents; each skill then expands into a small set of metric checks and diagnostics.

Intentional Change Verifier. This skill evaluates action-induced semantic changes. It asks whether the requested entity, attribute, relation, or event changed as intended, whether the correct target was changed, and whether unrelated variables were preserved. Representative checks include semantic accuracy, target accuracy, specificity, and preservation diagnostics. This separates successful intentional state updates from plausible clips that merely resemble the prompt.

Physical Response Verifier. This skill evaluates the first-order response to a physical intervention. It checks whether the target object or process visibly responds, whether motion concentrates in the intended region rather than the background, and whether optional visual-language judgement agrees that the response occurred. It is deliberately narrower than a full dynamics verifier: if the target never responds, more detailed physical-law or trajectory checks are not meaningful.

Physical Dynamics Verifier. This skill evaluates whether a physical process follows the constraints specified by the case. Depending on available evidence, it can use simulator references, analytic validators, trajectory constraints, process consistency checks, or task-specific kinematic validators. The skill is applicable only when the case construction provides enough structure to define the expected dynamics; otherwise physical plausibility and response checks are reported instead.

Drift Degradation Analyzer. This skill evaluates long-horizon degradation. It divides a rollout into chunks and tracks whether quality, identity, layout, or other invariant commitments degrade over time. Its output can include a chunk-wise curve, degradation drop, half-life, and diagnostic tags for identity or layout drift. This makes drift a measured persistence failure rather than an informal impression of video quality.

Return Consistency Verifier. This skill evaluates whether the generated world remains compatible after leaving and returning. It first checks that the rollout contains a valid return trigger, then compares entities, environment layout, and render consistency between the departure and return views. It can combine viewpoint evidence with semantic or visual evidence: the goal is not pixel-level reproduction, but compatibility with the same evolving world.

Offscreen Evolution Verifier. This skill evaluates processes that should continue while temporarily unseen. It checks whether the target reappears, whether its state has evolved in the expected direction, and whether relations to nearby objects remain continuous. Typical failure modes include freezing during occlusion, resetting when the object reappears, or jumping to an unrelated state. For physical processes with stronger references, this skill can be paired with the Physical Dynamics Verifier.

Skill Agent	Representative metrics / checks	Output
RENDER QUALITY INSPECTOR	Aesthetic quality, imaging quality, temporal flicker, motion smoothness, artifact and readability gates	Render quality score
PHYSICAL PLAUSIBILITY INSPECTOR	Support/contact, object integrity, scale and perspective consistency, motion feasibility, judgeability gate	Physical plausibility score
VIEWPOINT TRAJECTORY VERIFIER	Camera-path adherence, navigation consistency, layout/view recovery, return-trigger support	Viewpoint trajectory score
INTENTIONAL CHANGE VERIFIER	Semantic accuracy, target accuracy, specificity, preservation diagnostics	Intentional change score
PHYSICAL RESPONSE VERIFIER	Target motion energy, control-region motion, motion ratio, optional VLM response judgement	Physical response score
PHYSICAL DYNAMICS VERIFIER	Simulator or analytic validators, trajectory constraints, process consistency, task-specific kinematics	Physical dynamics score
DRIFT DEGRADATION ANALYZER	Chunk-wise quality curve, degradation drop, half-life, identity and layout drift diagnostics	Drift score
RETURN CONSISTENCY VERIFIER	Return trigger, entity consistency, environment consistency, render consistency	Return consistency score
OFFSCREEN EVOLUTION VERIFIER	Reappearance, expected evolution, relation continuity, freeze/reset diagnostics	Offscreen evolution score

Table 5: Representative metric composition of the current WorldProbe skill agents. The release implementation may report additional diagnostics, but the listed outputs are the score names consumed by WorldProbe aggregation.